$$z_G = z_0 \times \frac{z_1}{z_0} \times \frac{z_2}{z_1} \times \cdots \times \frac{z_m}{z_{m-1}}$$

$$= 2^M \cdot \varrho_1 \cdot \varrho_2 \cdots \varrho_m$$

where $\varrho_i = z_i / z_{i-1} \geq 1/2$ ensures efficient estimation